Divergence in a Collatz-Type Recursion

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Abstract

We study the map $T : \mathbb{N}_{\text{odd}} \to \mathbb{N}_{\text{odd}}$ defined by

$$T(n) = \frac{\lfloor \sqrt{2}\, n + 4 \rfloor}{2^{\nu_2(\lfloor \sqrt{2}\, n + 4 \rfloor)}}.$$

For every odd starting value n_0 the resulting sequence $(n_t)_{t\geq 0}$ diverges to infinity. More precisely, after a short initial segment, for every t either $n_{t+1} > n_t$ or $n_{t+2} > n_t$. Consequently, this provides a Collatz-type recursion with no finite limit cycles capturing all orbits.

Keywords Collatz problem; generalizations; divergence; counterexample.

1 Introduction

In 1937, Lothar Collatz studied the well-known recursion which, for odd n, maps $n \mapsto 3n+1$ and then repeatedly halves until an odd number is reached. The (still open) conjecture asserts that every n eventually enters the cycle $1 \to 4 \to 2 \to 1$; see [2]. The variant with $3n - 1$ leads to a different structure featuring three small limit cycles.

We consider a natural generalization: for real parameters (x, y) with $1 < x < 2$ and $y > 0$ and for odd n set

$$T_{x,y}(n) = \frac{\lfloor x\, n + y \rfloor}{2^{\nu_2(\lfloor x\, n + y \rfloor)}}, \tag{1}$$

so that one first takes an affine step and then divides out all powers of two. The choice $(x, y) = (\frac{3}{2}, 0)$ recovers the classical $3n - 1$ variant.

Our original conjecture was that for every pair (x, y) all odd starting values eventually enter one of finitely many cycles. In this note we present a counterexample to this conjecture for $(x, y) = (\sqrt{2}, 4)$: every odd orbit diverges.

2 Main result

Theorem 1. *For the recursion* $T(n) = \dfrac{\lfloor \sqrt{2}\, n + 4 \rfloor}{2^{\nu_2(\lfloor \sqrt{2}\, n + 4 \rfloor)}}$ *and every odd starting value* n_0*, the sequence* $(n_t)_{t\geq 0}$ *with* $n_{t+1} = T(n_t)$ *is unbounded. More precisely, there exists* $T = O(\log n_0)$ *such that for all* $t \geq T$ *either* $n_{t+1} > n_t$ *or* $n_{t+2} > n_t$*.*

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We set $\alpha = \sqrt{2}$ and write for odd n the decomposition

$$\alpha n = k(n) + x, \qquad k(n) = \lfloor \alpha n \rfloor \in \mathbb{Z},\ x \in (0,1), \tag{2}$$

which is unique since α is irrational. Define

$$R(n) := k(n) + 4, \qquad e(n) := \nu_2(R(n)), \qquad T(n) := \frac{R(n)}{2^{e(n)}}. \tag{3}$$

Lemma 1 (Transition of $\lfloor \alpha\, T(n) \rfloor$). *With notation as in (2)–(3) the following hold for odd n:*

1. *If $k(n)$ is odd (so $e(n) = 0$ and $T(n) = k(n) + 4$), then*

   $$\alpha\, T(n) = 2n - \alpha x + 4\alpha \in (2n + 3\alpha,\ 2n + 4\alpha) = (2n + 4.242\ldots,\ 2n + 5.656\ldots),$$

   hence $\lfloor \alpha\, T(n) \rfloor \in \{2n+4,\ 2n+5\}$ (in particular $\equiv 2, 3 \pmod 4$).

2. *If $k(n) \equiv 2 \pmod 4$ (so $e(n) = 1$ and $T(n) = (k(n)+4)/2$), then*

   $$\alpha\, T(n) = n - \tfrac{\alpha x}{2} + 2\alpha \in \left(n + 2\alpha - \tfrac{\alpha}{2},\ n + 2\alpha\right) = (n + 2.121\ldots,\ n + 2.828\ldots),$$

   hence $\lfloor \alpha\, T(n) \rfloor = n + 2$ (odd).

Proof. Substitute $\alpha n = k(n) + x$ into $\alpha T(n)$ and use $\alpha^2 = 2$. □

We now introduce the invariant

$$I(n): \quad k(n) = \lfloor \alpha n \rfloor \not\equiv 0 \pmod 4. \tag{4}$$

Lemma 2 (Persistence of the invariant). *Along the orbit starting at $n_0 = 1$, property $I(n_t)$ holds for all $t \geq 0$.*

Proof. Base: $k(1) = \lfloor \alpha \rfloor \equiv 1 \pmod 4$. Induction step: Suppose $I(n)$ holds. If $k(n)$ is odd, Lemma 1(1) gives $\lfloor \alpha T(n) \rfloor \in \{2n+4, 2n+5\}$, neither $\equiv 0 \pmod 4$. If $k(n) \equiv 2 \pmod 4$, then by Lemma 1(2) we have $\lfloor \alpha T(n) \rfloor = n + 2$ (odd). Thus $I(T(n))$ holds. □

The consequences for the number $e(n)$ of halvings in one step are immediate:

$$e(n) = \nu_2(k(n) + 4) = \begin{cases} 0, & k(n) \equiv 1, 3 \pmod 4, \\ 1, & k(n) \equiv 2 \pmod 4, \\ \geq 2, & k(n) \equiv 0 \pmod 4. \end{cases} \tag{5}$$

By Lemma 2, along the 1-orbit the last case never occurs, hence:

1. At most one halving per step.
2. No consecutive halving steps: if $e(n_t) = 1$, then by Lemma 1(2) $\lfloor \alpha n_{t+1} \rfloor = n_t + 2$ is odd, so $e(n_{t+1}) = 0$.

We next control the initial segment for arbitrary n_0 where multi-halving may occur.

Lemma 3 (End of multi-halving). *If $k(n_t) \equiv 0 \pmod 4$ (so $e(n_t) \geq 2$), then*

$$n_{t+1} = T(n_t) \leq \frac{\alpha n_t + 4}{4} = \frac{n_t}{\sqrt{8}} + 1.$$

Consequently, a run of consecutive steps with $k(n) \equiv 0 \pmod 4$ has length $O(\log n_0)$ and must terminate. In particular, there exists $T = O(\log n_0)$ such that for all $t \geq T$ one has $e(n_t) \in \{0, 1\}$.

Proof. Immediate from $T(n) = \frac{k(n)+4}{2^{e(n)}} \le \frac{\alpha n + 4}{4}$ when $e(n) \ge 2$, and a short induction. $\square$

Proof of Theorem 1. After the initial phase guaranteed by Lemma 3, only cases $e(n) \in \{0,1\}$ appear and never twice in a row with $e = 1$. If $e(n_t) = 0$ then $n_{t+1} = \lfloor \alpha n_t + 4 \rfloor > \alpha n_t > n_t$. If $e(n_t) = 1$ then

$$n_{t+2} > \alpha \frac{\lfloor \alpha n_t + 4 \rfloor}{2} \ge \alpha \frac{\alpha n_t + 3}{2} = n_t + \frac{3}{\sqrt{2}} > n_t + 2.$$

Thus for all large t, either $n_{t+1} > n_t$ or $n_{t+2} > n_t$. If (n_t) were bounded, some maximal value would recur infinitely often; the two-step drift precludes this, hence the sequence is unbounded. $\square$

3 Numerical evidence

For $n_0 = 1$ the first 30 steps are reported in Table 1. An asterisk marks a step with a single halving.

4 Discussion

(1) We expect that parameter pairs (x, y) in (1) that avoid finite attractors are rare; nevertheless, a general proof of convergence to finitely many limit cycles for a single explicit pair is still lacking.

(2) Without proof, integer-precise computations suggest that trajectories also diverge for $(x, y) = (\sqrt{2}, 6)$ with $n_0 = 1$, and $(x, y) = (\sqrt{2}, 24)$ with $n_0 = 1$, as well as $(x, y) = (\sqrt[3]{4}, 5.5)$ with $n_0 = 5$.

(3) For $(x, y) = (\frac{4}{3}, \frac{5}{2})$ we have a partial proof of divergence at $n_0 = 5$.

(4) The ideas herein may extend to other Collatz-type variants; cf. [1].

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References

[1] I. Althöfer, M. Hartisch, and T. Zipproth. Analysis of a Collatz Game and other variants of the $3n + 1$ Problem. ACG 2023, Springer Lecture Notes in Computer Science 14528 (2024), 123–132.

[2] J. Lagarias. The $3x + 1$ problem and its generalizations. *American Mathematical Monthly* 92 (1985), 3–23.

Table 1: Trajectory for $n_0 = 1$ up to $t = 30$. Asterisk "$*$" marks a halving step.

t	$n(t)$	halve?
0	1	
1	5	
2	11	
3	19	$*$
4	15	
5	25	
6	39	
7	59	
8	87	
9	127	
10	183	$*$
11	131	
12	189	
13	271	
14	387	
15	551	
16	783	
17	1111	
18	1575	
19	2231	
20	3159	
21	4471	$*$
22	3163	
23	4477	
24	6335	
25	8963	
26	12679	$*$
27	8967	
28	12685	
29	17943	
30	25379	